

Gate (II.11): scoping pass

Exact statement inventory for the interacting fixed-coarse limit,
the momentum-source obstruction, and the thermal two-torus route

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Abstract

This document is the preparatory scoping pass for the single remaining gate of the Part II programme: the interacting fixed-coarse characteristic-functional limit (II.11). It records, with proofs where they are short and load-bearing: (i) the *minimal* logical requirements of the gate — pointwise convergence in the coarse symbol suffices; no uniformity over source sets is required by the proved Weyl criterion; (ii) an exact obstruction which makes step (2) of the recorded attack plan (`strategy.md` §7) unimplementable as literally written: sharp-time momentum sources are not Cameron–Martin admissible for the Euclidean measures on either side (the relevant H^{-1} integral diverges identically), so “characteristic functionals including the momentum component” cannot be Gaussian source functionals; and (iii) a repaired architecture — the *thermal two-torus route* — in which both sides of (II.11) are represented by traces over the compact space–time torus $\mathbb{T}_t \times \mathbb{T}_L$, the momentum component enters through exact finite-mode Mehler formulas and a smooth mean shift rather than through distributional sources, the $t \rightarrow \infty$ and $N \rightarrow \infty$ limits are interchanged by a two-line spectral lemma, and the previously feared N -uniform gap input is *derived* (in eventual form) from partition-function convergence instead of being assumed. The route is reduced to eight statements A1–A8 with assigned proof methods; a fallback route B and its precise defect are recorded; the import-acquisition list and the numeric falsification checkpoint are specified. This document proves no part of (II.11); it fixes what “proving (II.11)” shall mean, statement by statement.

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1 The gate and its minimal logical requirements

Throughout, the model and notation are those of the Part II model sheet (`lampoort_part_ii_model_sheet.tex`): the spatial torus $\mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z})$; dyadic lattices Λ_N with spacing $\varepsilon_N = 2^{-N}\varepsilon_0$; Weyl algebras $\mathfrak{W}_{N,L}$ over $\ell^2(\Lambda_N)$ with the symplectic form of the model sheet; the $D4$ refinement embeddings $\alpha_M^N : \mathfrak{W}_{M,L} \rightarrow \mathfrak{W}_{N,L}$ and the continuum embeddings β_M ; the nearest-neighbour free Hamiltonian $H_{N,L}^{(0)}$ with dispersion $\gamma_N(k) = (m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2))^{1/2}$, $k \in \Gamma_N$; the finite-torus Wick covariance $c_{N,L}$; the interacting operator

$$K_{N,L} := H_{N,L}^{(0)} + \lambda \varepsilon_N \sum_{x \in \Lambda_N} : \Phi_N(x)^4 :_{c_{N,L}} \quad (g_N \equiv 1),$$

realized by its coercive closed quadratic form, with compact resolvent and simple ground state Ω_N , and the states $\omega_N := \omega_{N,L,\lambda,1} = \langle \Omega_N, \cdot \Omega_N \rangle$.

Definition 1.1 (the gate). Gate (II.11) is the statement: there is a state $\omega_{L,\lambda}^{\text{ct}}$ on the continuum Weyl algebra $\mathfrak{W}(\mathbb{T}_L)$ — namely the vector state of the ground vector of the operator K_L constructed in V3, Theorem 4.1 as the strong limit of the mode-truncated semigroups, in the convention $(a, b) = (0, 0)$, the plain φ^4 truncation convention (V2, Corollary 5.2; V3, §1) — such that for every fixed coarse scale M and every $\xi \in h_{M,L}$,

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))). \quad (\text{II.11})$$

Whether this K_L coincides with any *other* realization of $H_{0,L} + \lambda \int_{\mathbb{T}_L} : \varphi^4 :$ on $\mathbb{T}_L$ — in particular with a Segal essentially-self-adjoint one — is neither proved nor consumed anywhere in Part II; see V3, Remark 4.2, which records that essential self-adjointness of $(H_0 + V_L)|_{\mathcal{D}}$ is not established here and that $\omega_{L,\lambda}^{\text{ct}}$ is *defined* by this K_L . The gate is therefore an identification with a constructed object, not a uniqueness claim.

Remark 1.2 (what the Weyl criterion already delivers; demotion of uniformity). The full-sequence Weyl criterion and the identification corollary are *proved* (`lampoort_part_ii_weyl_reduction.tex`, Thm. 1.1 and Cor. 3.1): if the limit in (II.11) exists for every fixed M and ξ , then the limits define states ρ_M on each $\mathfrak{W}_{M,L}$, convergence upgrades automatically from Weyl generators to every $A \in \mathfrak{W}_{M,L}$ (a 2δ argument on the norm-dense span), the family is projectively consistent, and it assembles to the unique inductive-limit state; if moreover the limit equals the right side of (II.11) generator by generator, the inductive-limit state is $\omega_{L,\lambda}^{\text{ct}} \circ \beta$. The hypotheses are *pointwise in ξ* . Consequently:

1. Step (3) of the recorded attack plan (`strategy.md` §7: “prove convergence of those scalar functionals uniformly on a stated bounded set of source strengths”) demands more than the gate needs. Uniformity on bounded symbol sets is a *desirable output* of the estimates below (and would feed any later quantitative rate), but it is not a logical prerequisite and shall not be treated as one.
2. No positivity, normalization, or consistency statement needs separate proof at the gate: all are downstream of pointwise convergence, by the cited criterion.

Remark 1.3 (what is off the critical path — reaffirmed with one refinement). The corrected programme records that an N -uniform interacting gap, N -uniform local moments, and an interacting Lieb–Robinson bound are not on the static critical path. The scoping pass refines

this: the route below *derives* an eventual uniform gap (Statement A7, via Lemma 3.2) from partition-function convergence. It is an output, not an input — the same logical pattern as the Part I slab limit, where the gap of the cutoff theory was an output of CE2 and simplicity was the only spectral input.

2 The momentum-source obstruction and the design amendment

Step (2) of the recorded attack plan asks to “represent the two sides of (II.11) by their corresponding finite-torus Euclidean characteristic functionals, including the momentum component of ξ .” The following computation shows that the momentum component admits *no* representation as a Gaussian source functional, on either side, so step (2) is unimplementable as written and must be amended.

Lemma 2.1 (sharp-time momentum sources are not H^{-1}). *Let $C_\infty = (-\partial_t^2 - \partial_x^2 + m^2)^{-1}$ be the free Euclidean covariance on $\mathbb{R}_t \times \mathbb{T}_L$, and for real $h \in C^\infty(\mathbb{T}_L)$, $h \neq 0$, let $\delta'_0 \otimes h$ be the time-derivative sharp-time source, i.e. the functional $\varphi \mapsto -\partial_t \Phi(\delta_0 \otimes h)$ in the distributional sense. Then*

$$\|\delta'_0 \otimes h\|_{H^{-1}}^2 = \frac{1}{2L} \sum_{k \in \Gamma_\infty} |\hat{h}(k)|^2 \int_{\mathbb{R}} \frac{dk_0}{2\pi} \frac{k_0^2}{k_0^2 + \gamma(k)^2} = +\infty, \quad (1)$$

where $\gamma(k)^2 = k^2 + m^2$. The same divergence, with γ_N in place of γ , occurs for the Euclidean measure of the spatial-lattice model (whose time variable is continuous). Consequently the random variable “ $\Phi(\delta'_0 \otimes h)$ ” is not an element of the Gaussian first chaos on either side, and $e^{i\Pi(h)}$ -insertions cannot be written as $e^{i\Phi(J)}$ with an H^{-1} source J .

Proof. The Fourier symbol of $\delta'_0 \otimes h$ is $ik_0 \hat{h}(k)$, so the H^{-1} -norm is the displayed double sum; the k_0 -integrand tends to 1 as $|k_0| \rightarrow \infty$ for every fixed k , so already a single k -term with $\hat{h}(k) \neq 0$ diverges. For the lattice model the spatial sum is finite but the k_0 -integral is unchanged. $\square$

Remark 2.2 (design amendment D- α). The momentum component of $\xi = \varepsilon_M q + ip$ shall never enter path space as a source. It enters in exactly two controlled ways:

1. *Free/thermal level (exact formulas).* All free thermal Weyl expectations are computed in closed form by finite-mode Mehler algebra (Statement A3); no path-space representation is used for them.
2. *Interacting level (mean shift).* In the trace representation of Statement A2, the unitary $e^{i\Pi_N(p)}$ acts by translation of the field configuration at the marked time slice. The resulting “jump” ensemble is the translate of the periodic Gaussian ensemble by the *explicit mean path* $H_p(t')$ — the free covariance’s harmonic interpolation of the jump around the time circle — which is smooth away from the marked slice and lies in every L^p of the circle. The interaction then changes by the Wick-binomial finite sum

$$\mathcal{U}(\varphi + H_p) - \mathcal{U}(\varphi) = \lambda \int \left[4H_p : \varphi^3 : + 6H_p^2 : \varphi^2 : + 4H_p^3 \varphi + H_p^4 \right], \quad (2)$$

a polynomial perturbation with smooth deterministic coefficients, uniformly controlled in N because $H_p^{(N)} \rightarrow H_p$ by the dispersion estimates of the free-rate document. No distributional source ever appears. The precise bookkeeping is Statement A2’s content and is scheduled, not assumed.

This amendment replaces step (2) of `strategy.md` §7 and is recorded there.

3 Route A: the thermal two-torus architecture

3.1 The representation and the two elementary lemmas

For $t > 0$ define the thermal functionals

$$F_N(t; \xi) := \frac{\text{Tr}(e^{-tK_{N,L}} W_N(R_M^N \xi))}{\text{Tr}(e^{-tK_{N,L}})}, \quad F_\infty(t; \xi) := \frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr}(e^{-tK_L})}, \quad (3)$$

the right-hand version conditional on the continuum package A1 (trace-class semigroup). Both numerators are Euclidean objects over the *compact* space-time torus $\mathbb{T}_t \times \mathbb{T}_L$ (time circle of circumference t): no slab limit, no boundary terms, no spatial cutoff. The architecture rests on the following two lemmas, which we prove now because the entire route stands on them.

Lemma 3.1 (interchange of $t \rightarrow \infty$ and $N \rightarrow \infty$). *Let $(a_N(t))_{N \in \mathbb{N} \cup \{\infty\}}$, $t \geq t_0 > 0$, be complex functions with $|a_N(t)| \leq c_0$ for all N, t , and let ℓ_N be numbers such that*

$$|a_N(t) - \ell_N| \leq c_0 D_N(t), \quad D_N(t) := \sum_{j \geq 1} e^{-t(E_j(N) - E_0(N))}, \quad (4)$$

for discrete sequences $E_0(N) < E_1(N) \leq E_2(N) \leq \dots \rightarrow \infty$. Assume

- (i) $a_N(t) \rightarrow a_\infty(t)$ as $N \rightarrow \infty$, for every fixed $t \geq t_0$;
- (ii) $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ as $N \rightarrow \infty$, for one fixed $t_1 \geq t_0$;
- (iii) $\liminf_N (E_1(N) - E_0(N)) \geq \gamma > 0$.

Then $\ell_N \rightarrow \ell_\infty$.

Proof. For $t \geq t_1$,

$$D_N(t) = \sum_{j \geq 1} e^{-(t-t_1)(E_j(N) - E_0(N))} e^{-t_1(E_j(N) - E_0(N))} \leq e^{-(t-t_1)(E_1(N) - E_0(N))} D_N(t_1).$$

By (ii) there are N_0 and $D_* < \infty$ with $D_N(t_1) \leq D_*$ for $N_0 \leq N < \infty$ and also $D_\infty(t_1) \leq D_*$, and by (iii) there is $N_1 \geq N_0$ with $E_1(N) - E_0(N) \geq \gamma/2$ for $N_1 \leq N < \infty$. Hence

$$\sup_{N_1 \leq N < \infty} D_N(t) \leq D_* e^{-(t-t_1)\gamma/2} \xrightarrow[t \rightarrow \infty]{} 0.$$

Hypothesis (iii) is a statement about finite N only and says nothing about the limit member, which is handled separately by the standing strict inequality $E_0(\infty) < E_1(\infty)$ of the setup: $D_\infty(t) \leq e^{-(t-t_1)(E_1(\infty) - E_0(\infty))} D_\infty(t_1) \rightarrow 0$ as $t \rightarrow \infty$ as well. Given $\varepsilon > 0$ choose t with $\sup_{N_1 \leq N < \infty} D_N(t) < \varepsilon$ and $D_\infty(t) < \varepsilon$; then for $N \geq N_1$,

$$|\ell_N - \ell_\infty| \leq |\ell_N - a_N(t)| + |a_N(t) - a_\infty(t)| + |a_\infty(t) - \ell_\infty| \leq 2c_0\varepsilon + |a_N(t) - a_\infty(t)|,$$

and the middle term tends to 0 by (i). Thus $\limsup_N |\ell_N - \ell_\infty| \leq 2c_0\varepsilon$ for every ε . $\square$

Lemma 3.2 (Laplace transfer of spectra). *For $N \in \mathbb{N} \cup \{\infty\}$ let $\nu_N = \sum_j \delta_{E_j(N)}$ be the eigenvalue point measures (with multiplicity) of self-adjoint operators with $Z_N(t) := \int e^{-tE} d\nu_N(E) < \infty$ for all $t > 0$. Assume*

- (i) $Z_N(t) \rightarrow Z_\infty(t)$ as $N \rightarrow \infty$, for every $t \geq t_*$, where $t_* > 0$ is fixed but arbitrary;

(ii) $\inf_N E_0(N) > -\infty$.

Then $\nu_N \rightarrow \nu_\infty$ vaguely on $\mathbb{R}$; consequently, for every j , $E_j(N) \rightarrow E_j(\infty)$ (eigenvalues in increasing order with multiplicity), and if $E_0(\infty)$ is simple with gap $\gamma_L := E_1(\infty) - E_0(\infty) > 0$, then

$$E_1(N) - E_0(N) \rightarrow \gamma_L, \quad D_N(t) = e^{tE_0(N)} Z_N(t) - 1 \rightarrow D_\infty(t) \quad (t \geq t_*).$$

Proof. Let $c := \inf_N E_0(N)$. All ν_N are supported in $[c, \infty)$. The linear span $\mathcal{A}_{t_*}$ of $\{E \mapsto e^{-tE} : t \geq t_*\}$ is a subalgebra of $C_0([c, \infty))$ (indeed $e^{-tE}e^{-sE} = e^{-(t+s)E}$ with $t+s \geq 2t_* \geq t_*$) which separates points and vanishes nowhere, so by the Stone–Weierstrass theorem for locally compact spaces $\mathcal{A}_{t_*}$ is uniformly dense in $C_0([c, \infty))$. For $f \in \mathcal{A}_{t_*}$, (i) gives $\int f d\nu_N \rightarrow \int f d\nu_\infty$.

The passage to general f must be made against a *finite* measure: ν_N has infinite total mass, so $\|f - g\|_\infty < \varepsilon$ by itself controls nothing. (For $E_j(N) = j$, $f(E) = \varepsilon/(1+E) \in C_0$ and $g = 0$ one has $\int f d\nu_N = +\infty$.) We therefore pair against $e^{-t_*E} d\nu_N$, whose total mass is $Z_N(t_*)$ and hence uniformly bounded by (i). Let $f \in C_c(\mathbb{R})$ and $\varepsilon > 0$. Then $\tilde{f}(E) := f(E)e^{t_*E} \in C_c([c, \infty)) \subset C_0([c, \infty))$; pick $g \in \mathcal{A}_{t_*}$ with $\|f - g\|_\infty < \varepsilon$. Since $\int e^{-t_*E} d\nu_N = Z_N(t_*)$,

$$\left| \int f d\nu_N - \int g e^{-t_*E} d\nu_N \right| = \left| \int (\tilde{f} - g) e^{-t_*E} d\nu_N \right| \leq \varepsilon Z_N(t_*),$$

and likewise for ν_∞ . Moreover $g e^{-t_*E} \in \mathcal{A}_{t_*}$, so $\int g e^{-t_*E} d\nu_N \rightarrow \int g e^{-t_*E} d\nu_\infty$ by (i). Letting $\varepsilon \downarrow 0$ gives $\int f d\nu_N \rightarrow \int f d\nu_\infty$ for every $f \in C_c(\mathbb{R})$, which is vague convergence.

The restriction $t \geq t_*$ in the last conclusion is necessary, not cosmetic: hypothesis (i) says nothing about Z_N on $(0, t_*)$. Take $t_* = 1$, $\nu_\infty = \sum_{j \geq 0} \delta_j$ and $\nu_N = \nu_\infty + m_N \delta_N$ with $m_N := \lceil e^N/N \rceil$. Then $Z_N(t) < \infty$ for every $t > 0$; for $t \geq 1$ one has $m_N e^{-tN} \leq (e^N/N + 1)e^{-N} \rightarrow 0$, so (i) holds on $[t_*, \infty)$; $E_0(N) = 0$, so (ii) holds; and $E_j(N) = j \rightarrow E_j(\infty)$, so the eigenvalue and gap conclusions hold. Yet $D_N(t) - D_\infty(t) = m_N e^{-tN}$, which at $t = \frac{1}{2}$ is $\sim e^{N/2}/N \rightarrow \infty$. Only $t \geq t_*$ is claimed, and only one value $t_1 \geq t_*$ is consumed downstream (V4, Theorem 5.2(c)).

Vague convergence of integer-valued atomic measures to an atomic limit gives convergence of the ordered atoms with multiplicity on every compact interval whose endpoints are not atoms of ν_∞ ; since ν_∞ is locally finite and its atoms are discrete, $E_j(N) \rightarrow E_j(\infty)$ for every j . The gap statement is the case $j = 0, 1$; the convergence of $D_N(t) = e^{tE_0(N)} Z_N(t) - 1$ follows from $E_0(N) \rightarrow E_0(\infty)$ and (i). $\square$

Remark 3.3. Lemma 3.1 with $a_N = F_N(\cdot; \xi)$, $\ell_N = \omega_N(W)$, $c_0 = 2$, and the spectral tail bound $|F_N(t; \xi) - \omega_N(W_N(R_M^N \xi))| = |Z_N(t)^{-1} \sum_{j \geq 1} e^{-tE_j(N)} (\langle \phi_j, W \phi_j \rangle - \omega_N(W))| \leq 2D_N(t)/(1 + D_N(t)) \leq 2D_N(t)$ reduces (II.11) to: fixed- t convergence of the thermal functionals (A6), fixed- t convergence of partition functions feeding Lemma 3.2 (A7), and the continuum package (A1). This is the entire logical skeleton of Route A; everything else below is the analytic content of those three items.

3.2 Statement inventory A1–A8

Statement 3.4 (A1: continuum torus package). [*import + exec*] For $\lambda \geq 0$ and the convention of A4b: the continuum operator $K_L = H_{0,L} + \lambda \int_{\mathbb{T}_L} : \varphi^4 :_{C_L} dx$ is self-adjoint and bounded below (form sum); e^{-tK_L} is trace class for every $t > 0$ (hence discrete spectrum, $Z_\infty(t) < \infty$); the ground state is simple with gap $\gamma_L > 0$; and $\omega_{L,\lambda}^{\text{ct}}$ is its vector state on $\mathfrak{W}(\mathbb{T}_L)$. Trace-class of the free part is elementary ($\text{Tr } e^{-tH_{0,L}} = \prod_{k \in \Gamma_\infty} (1 - e^{-t\gamma(k)})^{-1} < \infty$

since $\sum_k e^{-t\gamma(k)} < \infty$; the interacting statements are classical finite-volume $P(\varphi)_2$ material. *Import candidates*: Høegh-Krohn's finite-temperature $P(\varphi)_2$ paper (CMP 38 (1974) 195–224) and Figari–Høegh-Krohn–Nappi (CMP 44 (1975) 265–278, $P(\varphi)_2$ on the de Sitter cylinder = spatial circle), plus the Q -space semiboundedness/hypercontractivity package of Simon's book as an import candidate. *Fallback*: prove from scratch on the torus by the Part I Appendix-C pattern (Nelson bounds on compact space are strictly easier than on $\mathbb{R}$); simplicity by positivity improvement; trace-class by Golden–Thompson against $e^{-tH_{0,L}}$ plus $e^{-\mathcal{U}} \in L^p$. *Deliverable*: a source-scope audit in the Part II dossier style deciding import vs. proof, with page images for whatever is imported. *Executed resolution (16 August 2026)*: V3 takes the proof route. It derives the finite-mode Gaussian unitary, closed form, compact embedding, Mehler–Trotter kernel, cyclic trace, mean shift, periodic cutoff tails, and positivity argument explicitly. Its only printed import is the precisely enumerated $\Gamma(A)$ cluster from Simon, with the consumed pages imaged in the dossier; no unproved transplantation from a line model to the circle is used.

Statement 3.5 (A2: trace representations with a marked slice). *[exec]* (a) Lattice: for the finite-dimensional Schrödinger operator $K_{N,L}$ and $W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}$, the kernel identity

$$\mathrm{Tr}(e^{-tK_{N,L}} W_N(\varepsilon_N q + ip)) = e^{i\theta(q,p)} \int_{\mathbb{R}^{\Lambda_N}} e^{-tK_{N,L}}(\varphi + p, \varphi) e^{i\varepsilon_N \sum_x q(x)\varphi(x)} d\varphi,$$

with the explicit CCR phase θ , and its Feynman–Kac form: the periodic Ornstein–Uhlenbeck ensemble on $\mathbb{T}_t \times \Lambda_N$ with a configuration jump at the marked slice, equal to the translate of the periodic ensemble by the explicit mean path $H_p^{(N)}$, with the interaction shifted per (2). All objects are finite-dimensional; the proof is Trotter plus Gaussian bridge algebra. (b) Continuum: the same statement over $\mathbb{T}_t \times \mathbb{T}_L$, with the mean path $H_p(t')$ given by the periodized kernel $\cosh((\frac{t}{2} - |t'|)\gamma) / \sinh(\frac{t}{2}\gamma)$ -type mode multipliers applied to p ; proof by A1's trace-class semigroup, cylinder approximation, and the same bridge algebra mode by mode. *Contract*: part (b) is formulated so that its free case is Statement A3's formula, and no object beyond A1 is consumed.

Statement 3.6 (A3: exact free thermal functionals and a thermal fixed-coarse rate). *[exec]* (a) Closed forms: for the free lattice and continuum models,

$$F_N^{(0)}(t; \xi) = \exp\left[-\frac{1}{4} Q_N^{(t)}(\xi)\right],$$

$$Q_N^{(t)}(\xi) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} \coth\left(\frac{t\gamma_N(k)}{2}\right) \left(\gamma_N(k)^{-1} |\widehat{q}(k)|^2 + \gamma_N(k) |\widehat{p}(k)|^2\right),$$

and the analogous $Q_\infty^{(t)}$ with γ , obtained per mode from the harmonic-oscillator identity $\langle W(z) \rangle_t = \exp[-\frac{1}{2}|z|^2 \coth(t\gamma/2)]$ (finite-mode Mehler computation; to be derived, not imported). At $t \rightarrow \infty$, $\coth \rightarrow 1$ recovers the vacuum formulas of the free-rate document, eq. (1.4)/(1.8). (b) Thermal fixed-coarse rate: for fixed M and ξ , uniformly in $t \geq t_0$,

$$|Q_{M+r}^{(t)}(R_M^{M+r}\xi) - Q_\infty^{(t)}(R_M^\infty\xi)| \leq C 2^{-\theta r},$$

by the same envelope/filter-tail/dispersion decomposition as the vacuum rate theorem, using $1 \leq \coth(t\gamma/2) \leq \coth(t_0 m/2)$ and, for $t \geq t_0$, $\gamma \geq m$,

$$\coth(t\gamma/2) - 1 = \frac{2}{e^{t\gamma} - 1} \leq \frac{2}{e^{t_0\gamma} - 1} \leq \frac{2e^{-t_0\gamma}}{1 - e^{-t_0m}}$$

(the naive bound $2e^{-t_0\gamma}$ is *false* for every $t_0, \gamma > 0$, since $2/(e^x - 1) - 2e^{-x} = 2/[e^x(e^x - 1)] > 0$, and the ratio blows up like $1/(t_0\gamma)$ as $t_0\gamma \downarrow 0$; only the corrected form above is available). The thermal weights improve, never damage, the high-momentum tails. *Contract*: (b) reuses the proved Lemmas 2.1–3.1 of the free-rate document verbatim; only the weight bookkeeping is new.

Statement 3.7 (A4: interaction comparison on the space–time torus). [*executed form, repaired 16 August 2026*] Use one Hermitian Gaussian Fourier family on D_∞ . Off the Nyquist column set $G_N = G$; at $k_e = -\pi/\varepsilon_N$ set $G_N(k_0, k_e) = 2^{-1/2}[G(k_0, k_e) + G(k_0, -k_e)]$. This is the V2 same-noise coupling and makes the lattice field real. Its mixed covariance has a translation-invariant part with symbol $\hat{c}_\times = (\hat{c}_N \hat{c}_\infty)^{1/2}$ and a separately bounded physical-edge remainder. Then:

- (a) L^2 convergence: $\|\mathcal{U}_N - \mathcal{U}\|_{L^2(\mathbb{P})} \leq C(1+t)(1+L)\varepsilon_N[1 + \log(1/\varepsilon_N)]^{3/2}$ for $tL \geq 1$. The executed fourth-chaos identity is

$$\|X_4 - Y_4\|_2^2 = 4!(2Lt)^{-2}B_4 + \text{Al}_N - 2\mathcal{E}_N^{\text{edge}}.$$

Here telescoping the principal products gives the two explicit sums $T_1 \leq C(tL)^3 \varepsilon_N^2 \log_N^3$ (dispersion) and $T_2 \leq C(tL)^3 \varepsilon_N^2 \log_N^2$ (missing inputs), while the three possible lattice alias outputs satisfy $\text{Al}_N \leq CtL\varepsilon_N^2 \log_N^3$. These are the bounds proved from V2’s displayed F_3 convolution estimate and one-dimensional temporal and spatial sum–integral comparisons; no power-counting slogan is used as a proof. At the symmetric Nyquist edges, four momenta can conserve exactly (two positive and two negative); their missing contribution is

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{\sum k_{0,i}=0} \prod_i \hat{c}_\times(k_{0,i}, \pi/\varepsilon_N) \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5.$$

This term is retained in the executed V2 proof.

- (b) *Finite renormalization dictionary* (A4b): the model sheet fixes Wick ordering by the *lattice vacuum* constant $c_{N,L}$, while the continuum interaction is Wick-ordered by a continuum convention c_L^{ct} (to be fixed: vacuum mode sum with a stated truncation). The recombination identity $:\varphi^4:_{c_1} = :\varphi^4:_{c_2} + 6(c_2 - c_1): \varphi^2:_{c_2} + 3(c_2 - c_1)^2$ shows: if $\delta_N := c_{N,L} - c_{L,N}^{\text{ct}}$ (matched truncations) converges to δ_∞ , then — *if the two Wick powers were taken of the same field* — the lattice theory would converge to the continuum theory with polynomial $\varphi^4 - 6\delta_\infty \varphi^2 + 3\delta_\infty^2$ (note the sign, which follows from $c_2 - c_1 = -\delta_N$ in the displayed identity; the version with $+6\delta_\infty$ printed here originally was a slip). *Superseded* (12 Aug 2026): the executed form of this clause is V2, Corollary 5.2, and its outcome is that *no* residue survives: the lattice and continuum fields’ equal-point variances also differ by exactly δ_∞ , which cancels the excess Wick subtraction, so the gate’s continuum polynomial is plain φ^4 and the endpoint convention is $(a, b) = (0, 0)$. δ_∞ remains the correctly computed offset between the two tadpoles. *Task (as executed)*: compute δ_∞ exactly (mode sums; the model sheet’s $c_N = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1)$ already fixes the divergent part), and *define* $\omega_{L,\lambda}^{\text{ct}}$ in Definition 1.1 as the vacuum of the so-normalized continuum polynomial. This is a convention lock, exactly parallel to Part I’s Remark on sharp-time Wick ordering; (II.11) without it is under-specified.

- (c) *Mean-shift terms*: the same L^2 statements for each term of (2), with the deterministic coefficients $H_p^{(N)} \rightarrow H_p$ supplied by A2/A3 dispersion estimates. The executed proof

uses the fundamental band $B_N = \Gamma_N$ and representative $[q]_N$ modulo $2\pi/\varepsilon_N$. For chaos order $j = 1, 2, 3$ it distinguishes the folded lattice multiplier $K_{N,j}^{\text{fold}}$ from the exact-output multipliers $K_{\times,j}$ and $K_{\infty,j}$. Because the asymmetric-band mixed kernel need not be real on the Nyquist edge, V2 defines its scalar surrogate as the real part of the mixed pairing. After extending the lattice coefficient by zero, that real surrogate quadratic difference is exactly

$$\langle \delta b, K_{\infty,j} \delta b \rangle + \langle b_N, (K_{N,j}^{\text{fold}} - K_{\infty,j}) b_N \rangle - 2\Re \langle b_N, (K_{\times,j} - K_{\infty,j}) b_{\infty,B_N} \rangle - 2\Re A_{\times,j},$$

where $A_{\times,j}$ is the explicitly summed set of nonzero exact outputs $q = p + 2n\pi/\varepsilon_N$. Fundamental-band differences are $O(\varepsilon_N^2 \log_N^2)$, high outputs are $O(\varepsilon_N^2 \log_N^{j-1})$, and coefficient powers are controlled by a written convolution telescope. The physical edge remainder is estimated separately. There is no unnamed tail term. The resulting rate is $O(2^{-r(1+\eta)/4} + \varepsilon_N \log_N)$.

Superseded (13 Aug 2026): the coupling scheduled here — the image of the continuum space–time field under a sharp-time per-spatial-mode amplitude ratio — is *not* the coupling executed, and would not do what it is asked to do: a per-mode multiplier applied at a single time does not reproduce the lattice ensemble’s time correlations. What is executed is V2’s *same-noise* coupling (V2, Definition 1.1): one Hermitian Gaussian family G on D_∞ , with η built from the symbol $\hat{c}_\infty^{1/2}$ over D_∞ and η_N from $\hat{c}_N^{1/2}$ over D_N , the edge column $k_e = -\pi/\varepsilon_N$ symmetrised as $G_N(k_0, k_e) = 2^{-1/2}(G(k_0, k_e) + G(k_0, -k_e))$ (without which η_N is neither real nor of the periodic-lattice law). Clause (a) as executed reads $\|\mathcal{U}_N - \mathcal{U}\|_{L^2(\mathbb{P})} \leq C(1+t)(1+L)\varepsilon_N(1+\log(1/\varepsilon_N))^{3/2}$ under the standing hypothesis $tL \geq 1$ (V2, Theorem 4.3). Recorded as an erratum in V4, Remark 1.1.

Statement 3.8 (A5: uniform-in- N Nelson bounds on the space–time torus). *[exec]* For every $p < \infty$ and $t > 0$: $\sup_N \|e^{-\mathcal{U}_N}\|_{L^p(\mu_{t,L,N}^{\text{per}})} < \infty$, and the same for the mean-shifted interactions of (2) locally uniformly in p -shift. *Proof pattern:* verbatim the Part I Appendix C stability argument with the mollification scale κ replaced by ε_N^{-1} : the Wick lower bound $:P:_{\mathbf{c}} \geq -b(1+c)^{p'}$ with $c = c_{N,L} \sim \frac{1}{2\pi} \log \varepsilon_N^{-1}$, the L^2 ultraviolet difference between scales N and N' (part (a) computations), Nelson hypercontractivity on the Gaussian space, and the double-exponential tail with $2q = e^{L/(4p')}$. Nothing in that argument used noncompactness of space or vacuum boundary conditions; the periodic covariance obeys the same two Bessel-type bounds.

Amended (14 Aug 2026): as executed (V2, Theorem 7.2) the bound is stated for $t \in [t_0, T]$ under the standing convention $t_0 L \geq 1$, with a constant depending also on the coarse scale through ε_M ; the unrestricted range printed above is narrower in execution. The gate consumes A5 only at $t \geq t_* = \max(1, 1/L)$, so nothing is lost.

Statement 3.9 (A6: fixed- t convergence of the thermal functionals). *[exec]* For every fixed $t \geq t_0$, M , and ξ : $F_N(t; \xi) \rightarrow F_\infty(t; \xi)$. *Proof shape (all pieces above):* write both sides via A2 as $\mathbb{E}[e^{i\Phi(J_q^{(\bullet)})} e^{-\mathcal{U}_\bullet - \Delta \mathcal{U}_{\bullet,p}}] \cdot G_\bullet^{(0)}(t; \xi)$ normalized by the sourceless expectations; the free prefactors converge by A3; the exponents converge in every L^p ($p < \infty$) by A4 (a,c) plus Gaussian hypercontractivity; uniform integrability of $e^{-\mathcal{U}_N}$ by A5; conclude by Vitali (the same convergence pattern as Part I, Proposition on path regularity, Step 2). The q -source J_q is an honest H^{-1} element (sharp-time *position* sources are admissible: $\int dk_0 (k_0^2 + \gamma^2)^{-1} < \infty$), so no obstruction arises on the position side — Lemma 2.1 concerns momentum only.

Statement 3.10 (A7: partition-function convergence and the derived eventual gap). *[exec]* For every $t > 0$: $Z_N(t)/Z_N^{(0)}(t) \rightarrow Z_\infty(t)/Z_\infty^{(0)}(t)$ (this is A6 at $\xi = 0$, sourceless), and

$Z_N^{(0)}(t) \rightarrow Z_\infty^{(0)}(t)$ (explicit: $\prod_{k \in \Gamma_N} (2 \sinh(t\gamma_N(k)/2))^{-1} \rightarrow \prod_{k \in \Gamma_\infty} (2 \sinh(t\gamma(k)/2))^{-1}$, by the dispersion lemma and the summable tail $e^{-t\gamma(k)}$). *Amended (12 Aug 2026)*: the displayed free-factor limit is vacuous as written — both sides carry the divergent ground-energy factor and the right-hand infinite product tends to 0, since $\sum_k \gamma(k) = \infty$. A7 is executed in the *normal-ordered* normalization, where the free factor is $\prod_{k \in \Gamma_N} (1 - e^{-t\gamma_N(k)})^{-1} \rightarrow Z_0(t)$; see V4, Remark 1.1(2) and Lemma 5.1. Every quantity the route consumes (eigenvalue differences, D_N , gaps) is invariant under the c -number shift.

The conclusions must be read in that same normalization; in the *un*-shifted symbols two of them are false, since $e_0(N) = \frac{1}{2} \sum_{k \in \Gamma_N} \gamma_N(k) \geq r_N m \rightarrow \infty$ forces $Z_N(t) \rightarrow 0 \neq Z_\infty(t)$ and $E_j(N) \rightarrow +\infty$. Writing $\tilde{Z}_N(t) := \text{Tr } e^{-t(K_{N,L} - e_0(N))}$ and $\tilde{E}_j(N) := E_j(N) - e_0(N)$: for every $t \geq t_*$ ($t_* > 0$ fixed — see hypothesis (i) of Lemma 3.2 and V2, the remark following Theorem 4.3, the interaction comparison being unavailable as $t \downarrow 0$) one has $\tilde{Z}_N(t) \rightarrow Z_\infty(t) = \text{Tr } e^{-tK_L}$; with the uniform lower bound $\inf_N \tilde{E}_0(N) \geq -c$ (from A5 via $e^{-t\tilde{E}_0(N)} \leq \tilde{Z}_N(t) \leq \tilde{Z}_N^{(0)}(t) \|e^{-\mathcal{U}_N}\|$ -type bounds), Lemma 3.2 applied to $\tilde{\nu}_N = \sum_j \delta_{\tilde{E}_j(N)}$ yields $\tilde{E}_j(N) \rightarrow \mu_j$, the *derived* eventual uniform gap $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$, and $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ — the last two being invariant under the c -number shift, and being the only conclusions of A7 that A8 consumes.

Statement 3.11 (A8: assembly). *[exec, short]* A6 + A7 + A1 verify hypotheses (i)–(iii) of Lemma 3.1 for $a_N = F_N$, $\ell_N = \omega_N(W_N(R_M^N \xi))$, $\ell_\infty = \omega_{L,\lambda}^{\text{ct}}(W_{\text{ct}}(R_M^\infty \xi))$ (the spectral tail bound preceding §3.2 supplies (4) on both sides; the $N = \infty$ instance of (4) uses A1’s gap). Hence (II.11) holds pointwise for every M, ξ ; the proved Weyl criterion and identification corollary then deliver the full-sequence, identified, fixed-coarse theorem. No statement beyond A1–A7 is consumed.

3.3 What Route A does not need

No N -uniform gap as an input (A7 derives an eventual one); no interacting Lieb–Robinson estimate; no spatial cutoff (torus, $g \equiv 1$); no slab limits or boundary conditions (compact time circle); no uniformity of (II.11) in ξ (Remark 1.2); no comparison of the full lattice and continuum *measures* — Statement A4’s coupling compares only fourth-moment functionals, which is essential: the per-mode Gaussian relative densities diverge near the band edge (the thermal shadow of the finest-scale obstruction), so any route through measure-level absolute continuity would fail.

4 Route B (slab diagonal), recorded as fallback

The previously envisaged route represents ω_N by $T \rightarrow \infty$ slabs and compares slabs at fixed T . Its defect, stated precisely: the error $|\omega_N(W) - \text{slab}_N(T; W)|$ is controlled only by the *lattice* gap $\gamma_{N,L}$, so interchanging $T \rightarrow \infty$ with $N \rightarrow \infty$ requires exactly the uniformity that Route A derives in A7 — but without the trace structure, no partition-function argument is available to derive it, and one is thrown back on an N -uniform cluster-expansion input (the Battle–Federbush audit was negative). Route B is therefore strictly weaker than Route A at current knowledge and is kept only as a fallback should the trace-class package A1 fail to close (it will not: the free trace is elementary and the interacting dressing is A5).

5 Imports to acquire, and the numeric checkpoint

5.1 Acquisition list

For A1: Høegh-Krohn, *Relativistic quantum statistical mechanics in two-dimensional space-time* (CMP 38 (1974) 195–224); Figari–Høegh-Krohn–Nappi (CMP 44 (1975) 265–278). Both to be added to `refs/part_ii/` with page-image dossier entries if used as imports. Everything else in A2–A8 is self-contained or already proved in the Part II documents (free-rate lemmas) and Part I Appendix C (stability pattern).

5.2 Numeric falsification checkpoint (before execution)

A `numpy` script, in the style of `m2_ii_checkpoint.py`:

1. verify the A3 closed forms against exact diagonalization of small free lattice models (one and two modes, then $r_N \leq 64$), including the momentum component and the CCR phase of A2(a);
2. measure the thermal fixed-coarse rate exponent at $t \in \{1, 2, 4\}$ against the vacuum exponent θ (they should agree within the envelope constants);
3. for a two-site anharmonic toy (Λ of size 2, quartic coupling), verify the interchange mechanism numerically: $E_j(N)$ -convergence under mode-doubling, the $D_N(t)$ tail, and $F_N(t) \rightarrow \omega_N$ rates — a sanity check of Lemmas 3.1–3.2 in vivo;
4. compute $\delta_N = c_{N,L} - c_{L,N}^{\text{ct}}$ for the A4b dictionary numerically before the analytic evaluation, to fix the expected constant.

Falsification criteria: any mismatch in (1) kills A2/A3 as formulated; a t -dependent drift in (2) kills the claimed t -uniformity of A3(b); failure of eigenvalue convergence in (3) would signal a flaw in the A7 logic.

6 Execution phases and gates

phase	content	gate
V0	numeric checkpoint (§5.2)	all four criteria pass; exponents recorded
V1	A2 (trace representations) + A3 (free thermal formulas and rate)	proofs complete; A3(a) matches V0 item (1) to rounding
V2	A4 (interaction L^2 + dictionary δ_∞ ; four-edge, real-surrogate, and shifted fundamental-band repairs incorporated) + A5 (uniform Nelson)	constants explicit; δ_∞ matches V0 item (4)
V3	A1 (torus package)	full proof executed; finite-mode normalization and periodic tails explicit; gap $\gamma_L > 0$ certified
V4	A6 + A7 + A8 (assembly)	(II.11) closed; Weyl criterion invoked; adversarial pass on the assembled document

Standing rules carried over from Part I: transcription discipline for every display reused from the free-rate document and Appendix C; every import through a dossier entry with

page images; a claim ledger in the final document; an independent adversarial pass before the gate is declared closed.

7 Claim ledger of this scoping pass

claim	status	ground
Pointwise-in- ξ convergence suffices for the gate	proved elsewhere	Weyl-reduction document, Thm. 1.1/Cor. 3.1; Remark 1.2
Sharp-time momentum sources inadmissible	proved here	Lemma 2.1, display (1)
Interchange of limits under (i)–(iii)	proved here	Lemma 3.1
Spectral transfer from partition functions	proved here	Lemma 3.2
Free thermal formulas and thermal rate	scheduled [exec]	Statement A3; V0/V1
Trace representations with marked slice	scheduled [exec]	Statement A2; V1
Interaction L^2 comparison + renormalization dictionary	scheduled [exec]	Statement A4; V2
Uniform Nelson bounds (torus, all N)	scheduled [exec]	Statement A5 (Part I App. C pattern); V2
Continuum torus package	import/exec	Statement A1; V3
Fixed- t convergence; partition functions; assembly	scheduled [exec]	Statements A6–A8; V4
Eventual uniform lattice gap	derived, not assumed	A7 via Lemma 3.2
Measure-level absolute continuity route	rejected	band-edge divergence of per-mode relative densities (§3.3)
Route B (slab diagonal)	fallback only	§4